Form 63 game. 29 Apr. 2027 Reasoning by the absurd and by disjunction of cases

1. Warming · Flash [IVD] What is the parity of $n(n+1)$ for an integer n ?

Rep. Always even: one of the two consecutive integers is even.

2. Heart · Disjunction of cases [DEM] Show that $n(n+1)$ is even for the integer n.

Rep. If n is even, the product contains an even factor. n is odd, $n+1$ is even. In both cases, product is even.

3. Heart + · Absurd [DEM] Show by the absurd that the sum of a rational r and an irrational a is Irrational.

Rep. Suppose $r+a=q\in\mathbb{Q}$; then $a=q-r\in\mathbb{Q}$ (difference of two rationals): contradiction with a Irrational.

4. Back · Formula [IMG] $g(x)=2x+5$. Image of -1 ? Preimage of 11 ?

Rep. $g(-1)=3$; $2x+5=11\Leftrightarrow x=3$.

5. Red thread · Formula → graph [REG] Place the dots off -2 ; -1 ; 0 ; 1 ; 2 of the curve of $f(x)=x^2$; give their orders.

Rep. 4 ; 1 ; 0 ; 1 ; 4 : the points are symmetrical in relation to the axis of the ordered.

Form 64 Mon. 3 May 2027 Logical, contrapositive and counter-example connectors in situation

1. Warming · Flash [DEM] Write the denial of " $x>2$" and then " $x>2$ or $x<-1$ ».

Rep. $x\leqslant 2$; $x\leqslant 2$ and $x\geqslant -1$ or $-1\leqslant x\leqslant 2$.

2. Heart · Method [DEM] Write the contraposition of "if n^2 is even, so n is even".

Rep. "If n is odd, so n^2 is odd" (equivalent to the original statement).

3. Heart + · Counterexample [DEM] Refuse: "any prime number is odd."

Rep. Counter-example: 2 is first and even.

4. Return · True/false [PNP] True or false? Justify. The perimeter of a square is proportional to the length On his side.

Rep. Right. $P=4c$: Proportionality coefficient 4.

5. Red thread · Code reading [ALG] What's sent back $h(3)$? Write $h(x)$ in the form of an expression.

Rep. $h(3)=5\times 5=25$; $h(x)=(x+2)^2$.

Probabilities: Packing and weighted trees

Form 65 Mar. 4 May 2027 Probabilistic model and equiprobability

1. Warming · Flash [PRB] We're throwing a balanced piece three times.

Rep. $2^3=8$ equiprobable issues.

2. Heart · Method [PRB] Unfixed: $P(6)=0.25$ and the other five sides are equiprobable. $P(1)$.

Rep. $\dfrac{1-0{,}25}{5}=0{,}15$.

3. Heart + · Critical Spirit [PRB] "It will rain or it will not rain tomorrow, so there is one in two chances." Is this equiprobable model justified?

Rep. No: having two issues does not make them equiprobable. Equiprobability is a hypothesis of the model to be justified.

4. Back · Flash [Sighs] Which figures are composed of a pyramid pattern with a square base?

Rep. 1 square and 4 triangles (isoceles).

5. Red wire · Translate [LIT] A side square x See each side increase by 3. the increase in proportion to the x.

Rep. $(x+3)^2-x^2=6x+9$.

Form 66 Mon. 10 May 2027 Law of large numbers (vulgarized version): simulation

1. Warming · Flash [STA] Out of 100 launches, 47 "cells". What is the frequency of "cells"?

Rep. 0,47.

2. Heart · Interpret [PRB] Why, on 10 throws of a balanced piece, can the frequency of "cell" What's going on with 10,000 shots?

Rep. Sampling fluctuation: small series, high variability.
stabilizes near probability 0.5 (Large Numbers Act).

3. Heart + · True/false [PRB] True or false? Justify. By launching 1,000 times a balanced piece, we get exactly 500 "cells".

Rep. Wrong. We get a number close to 500, rarely exactly 500.

4. Back · Problem [MOD] Price incl. $P(x) = 1{,}2\,x$ (20% VAT). $P(250)$ and the preimage of 480.

Rep. $P(250) = 300$; $1{,}2x = 480 \Leftrightarrow x = 400$.

5. Red thread · Method [EQ] Solve in $\mathbb{R} : \dfrac{2x-1}{3} = \dfrac{x+4}{2}$.

Rep. $x = 14$
Cross products: $2(2x-1) = 3(x+4)$.

Form 67 March 11 May 2027 Python simulation of a repeated random experiment

1. Warming · Flash [PRB] $P(A) = 0.35$. Calculate $P(\overline{A})$.

Rep. $1 - 0{,}35 = 0{,}65$.

2. Heart · Code reading [ALG] What does this program calculate? What value will it display about?

```
from random import randint
account = 0
for i in range(1000):
        if randint(1, 6)==6:
                account = account + 1
print(account / 1000)
```

Rep. The frequency of appearance of 6 of 1 000 lances of a dice ; about $\dfrac{1}{6} \approx 0{,}167$.

3. Heart + · Method [PRB] Balanced di 12 faces numbered from 1 to 12. Likelihood of getting a number First?

Rep. First: 2 ; 3 ; 5 ; 7 ; 11 : $\dfrac{5}{12}$ (1 is not first).

4. Back · Construction programme [chuckles] M belongs to the mediator of $[AB]$. What can be said of MA and MB How can I build it in a compass?

Rep. $MA = MB$. Construction: two arcs of the same radius (greater than $\frac{1}{2}AB$) centered in A and B ; their intersections define the mediator.

5. Red thread · Recognize [SOL] What is the nature of the section of a cube by a plane parallel to one of the His faces?

Rep. A square on the same side as the face.

Form 68 13 May 2027 Conditional likelihood P $_A(B)$

1. Warming · Flash [PRB] Of the 25 half boarders, 15 are girls.
Like it's a girl?

Rep. $\dfrac{15}{25} = 0{,}6$.

2. Heart · Method [PRB] In a class of 30 students, 12 play football, 5 of which also play hand.
Choose a student at random: calculate $P_F(H)$, F : "Play football", H : "Play the Hand".

Rep. $P_F(H) = \dfrac{\text{Card}(F \cap H)}{\text{Card}(F)} = \dfrac{5}{12}$.

3. Heart + · Method [PRB] In an equiprobable model, $\text{Card}(A) = 50$ and $\text{Card}(A \cap B) = 20$. Calculate $P_A(B)$.

Rep. $\frac{20}{50} = 0{,}4$.

4. Back · Flash [NUM] Store in ascending order: $\frac{7}{4}, -\frac{3}{2}, 0{,}5$.

Rep. $-\frac{3}{2} < 0{,}5 < \frac{7}{4}$ $(-1{,}5 < 0{,}5 < 1{,}75)$.

5. Red thread · True/false [DEM] True or false? Justify. A quadrilateral with two opposite sides parallel is a parallelogram.

Rep. False. Counter-example: a trapeze (non-parallelogram) has only two parallel sides.

Form 69 March 18 May 2027 Conditional likelihood and cross-table

1. Warming · Flash [PRB] In the following table, what is the total of the "sport" line?

	Girls	Boys	Total
Sport	18	22	…
No sports	12	8	20
Total	30	30	60

Rep. $18 + 22 = 40$.

2. Heart · Method [PRB] We pick a student at random from this table. $P(S)$, $P_F(S)$ and $P_S(F)$ (F : girl, S : sport).

Rep. $P(S) = \frac{40}{60} = \frac{2}{3}$; $P_F(S) = \frac{18}{30} = 0{,}6$; $P_S(F) = \frac{18}{40} = 0{,}45$.

3. Heart + · True/false [PRB] True or false? Justify. $P_F(S) = P_S(F)$.

Rep. False. $0.6 \neq 0.45$: Reference sets (30 girls and 40 athletes) differ.

4. Back · Flash [Sighs] Calculate $5^0 + 2^{-1}$.

Rep. $1 + \frac{1}{2} = \frac{3}{2}$.

5. Red thread · Method [REP] $ABCD$ is a parallelogram with $A(1;1)$, $B(5;2)$, $C(6;5)$. Determine D.

Rep. The diagonals are even midpoint: $D = A + C - B = (2;4)$. Control: $\overrightarrow{AB} = \overrightarrow{DC} = (4;1)$.

Form 70 game. 20 May 2027 Weighted trees: construction

1. Warming · Flash [PRB] What is the value of the sum of the probabilities of branches from the same node?

Rep. 1.

2. Heart · Method [PRB] In a population, 30% have a disease M. A test is positive (T) for 90% of patients and 5% of non-malads.
I'm a tree.

Rep. $P(M) = 0{,}3$, $P(\overline{M}) = 0{,}7$; $P_M(T) = 0{,}9$, $P_M(\overline{T}) = 0{,}1$; $P_{\overline{M}}(T) = 0{,}05$, $P_{\overline{M}}(\overline{T}) = 0{,}95$.

3. Heart + · Translate [PRB] Translate into symbols: " 90% of patients have a positive test".

Rep. $P_M(T) = 0.9$: branch $M \to T$; do not write $P(T) = 0{,}9$.

4. Back · Flash [CAL] Calculate $\frac{2}{3} \div \frac{4}{9}$.

Rep. $\frac{2}{3} \times \frac{9}{4} = \frac{18}{12} = \frac{3}{2}$.

5. Red thread · Problem [IVD] Three headlights flash every 6, 8 and 9 seconds. $t = 0$. When the next simultaneous flashing?

Rep. PPCM $(6\,;\,8\,;\,9) = 72$: after 72 seconds.

Form 71 Mon. 24 May 2027 Weighted trees: multiplication along a path

1. Warming · Flash [CAL] Calculate $0.3 \times 0{,}9$.

Rep. $0{,}27$.

2. Heart · Method [PRB] With the tree of the sheet 70, calculate $P(M \cap T)$ and $P(\overline{M} \cap T)$.

Rep. $P(M \cap T) = 0{,}3 \times 0{,}9 = 0{,}27$; $P(\overline{M} \cap T) = 0{,}7 \times 0{,}05 = 0{,}035$.

Multiply along a path: P $(M \cap T) = P(M) \times P_M(T)$.

3. Heart + · Method [PRB] Deduct from it $P(T)$.

Rep. Two paths lead to T : $0{,}27 + 0{,}035 = 0{,}305$.

4. Return · Factorize [DEV] Factoring $x^2 - 9$.

Rep. $(x-3)(x+3)$.

5. Red thread · Flash [GEO] The side $[BC]$ of length 13 cm is hypotenuse of a right triangle ABC. What's the radius of his circumscribed circle?

Rep. The midpoint of hypotenuse is the center: radius $= 13 \div 2 = 6.5$ cm.

Form 72 March 25 May 2027 From painting to tree and symbolic writing

1. Warming · Flash [PRB] Express $\dfrac{30}{60}$ in the form of probability.

Rep. $0{,}5$.

2. Heart · Method [PRB] From the table on sheet 69, build the tree (F or Gthen S or $\overline{S}$) : give $P(F)$, $P_F(S)$ and $P_G(S)$.

Rep. $P(F) = 0{,}5$; $P_F(S) = \dfrac{18}{30} = 0{,}6$; $P_G(S) = \dfrac{22}{30} = \dfrac{11}{15}$.

3. Heart + · Translate [PRB] Write symbolically and calculate "the probability of being a boy who does not no sport".

Rep. $P(G \cap \overline{S}) = \dfrac{8}{60} = \dfrac{2}{15}$; or $0.5 \times \dfrac{8}{30}$.

4. Back · Flash [QP] A car runs at constant speed: 150 km in 2 hours.How far in 5 hours?

Rep. $75 \times 5 = 375$ km.

5. Red thread · Statement → formula [REG] A taxi charges 2,000 Ar of support and then 1,500 Ar per km. Express Price $p(x)$ for x km, then calculate $p(8)$.

Rep. $p(x) = 1\,500x + 2\,000$; $p(8) = 14{,}000$ Ar.

Form 73 27 May 2027 Inversion of conditioning: P $_A(B)$ *et* $P_B(A)$

1. Warming · Flash [PRB] What a difference between $P_M(T)$ and $P_T(M)$?

Rep. $P_M(T)$: probability of being positive knowing that you are sick; $P_T(M)$: to be sick knowing that we are positive.

2. Heart · Method [PRB] With data from form 70 ($P(M \cap T) = 0.27$ and $P(T) = 0.305$), calculate $P_T(M)$.

Rep. $P_T(M) = \dfrac{P(M \cap T)}{P(T)} = \dfrac{0{,}27}{0{,}305} \approx 0{,}885$.

3. Heart + · Method [PRB] Rare disease (1% of patients); positive test for 99% of patients and 5% of patients No malady. $P_T(M)$.

Rep. $P(M \cap T) = 0{,}0099$; $P(\overline{M} \cap T) = 0{,}99 \times 0{,}05 = 0{,}0495$; $P(T) = 0{,}0594$; $P_T(M) = \dfrac{0{,}0099}{0{,}0594} = \dfrac{1}{6} \approx 0{,}17$.

Counter-intuitive result: a "reliable to 99 % It is not enough.

4. Back · Graph [IMG] Curve of f :